

1. จงแสดงวิธีทำและหาดีเทอร์มิแนนต์ของ

$$A = \begin{bmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & -3 \\ 1 & 1 & 3 & 2 & 0 \end{bmatrix}$$

$$\begin{aligned} |A| &= \cancel{0}(C_{11}) + 1(C_{12}) + \cancel{0}(C_{13}) + (-1)(C_{14}) + 1(C_{15}) \\ &= 0 + 13 + 0 - 115 - 33 \\ &= -135 \end{aligned}$$

$$C_{22} = \begin{bmatrix} 2 & 1 & 3 & -2 \\ 1 & -1 & 2 & 3 \\ 3 & 2 & 4 & -3 \\ 1 & 3 & 2 & 0 \end{bmatrix} \quad \begin{aligned} |C_{22}| &= -2(C_{14}) + (3)(C_{24}) + (-3)(C_{34}) + 0(C_{44}) \\ &= (-2)(-8) + (3)(3) + (-3)(4) + 0 \\ &= 16 + 9 - 12 \\ &= 13 \end{aligned}$$

$$C_{14} = - \begin{bmatrix} 1 & -1 & 2 \\ 3 & 2 & 4 \\ 1 & 3 & 2 \end{bmatrix} = -8$$

$$C_{24} = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & 2 \end{bmatrix}$$

$$= 3$$

$$C_{34} = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 2 \\ 1 & 3 & 2 \end{bmatrix}$$

$$= 4$$

$$C_{42} = \begin{bmatrix} 2 & 1 & 3 & -2 \\ -2 & 3 & 2 & -1 \\ 1 & -1 & 2 & 3 \\ 1 & 3 & 2 & 0 \end{bmatrix} \quad |C_{42}| = (-2)(C_{14}) + (-1)(C_{24}) + (3)(C_{34}) + (0)(C_{44})$$

$$= (-2)(-24) + (-1)(-4) + (3)(21)$$

$$C_{14} = - \begin{bmatrix} -2 & 3 & 2 \\ 1 & -1 & 2 \\ 1 & 3 & 2 \end{bmatrix} = 48 + 4 + 63$$

$$= -24$$

$$C_{24} = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 2 \\ 1 & 3 & 2 \end{bmatrix}$$

$$= -4$$

$$C_{34} = - \begin{bmatrix} 2 & 1 & 3 \\ -2 & 3 & 2 \\ 1 & 3 & 2 \end{bmatrix}$$

$$= 21$$

$$C_{52} = - \begin{bmatrix} 2 & 1 & 3 & -2 \\ -2 & 3 & 2 & -1 \\ 1 & -1 & 2 & 3 \\ 3 & 2 & 4 & -3 \end{bmatrix} \quad |C_{52}| = (1)(C_{31}) + (-1)(C_{32}) + (2)(C_{33}) + (3)(C_{34})$$

$$= 3 + (-1)(3) + (2)(3) + (3)(9)$$

$$= -(3 - 3 + 6 + 27)$$

$$= -33$$

$$C_{31} = \begin{bmatrix} 1 & 3 & -2 \\ 3 & 2 & -1 \\ 2 & 4 & -3 \end{bmatrix}$$

$$= 3$$

$$C_{32} = - \begin{bmatrix} 2 & 3 & -2 \\ -2 & 2 & -1 \\ 3 & 4 & -3 \end{bmatrix}$$

$$= 3$$

$$C_{33} = \begin{bmatrix} 2 & 1 & -2 \\ -2 & 3 & -1 \\ 3 & 2 & -3 \end{bmatrix}$$

$$= 3$$

$$C_{34} = - \begin{bmatrix} 2 & 1 & 3 \\ -2 & 3 & 2 \\ 3 & 2 & 4 \end{bmatrix}$$

$$= 9$$

2. จงแสดงวิธีทำและหาค่า $|A^{-1}|$

นั่นคือ เริ่มต้นด้วยการหา A^{-1} จากนั้นให้ทำการหาดีเทอร์มิแนนต์ของเมทริกซ์นั้น

$$A = \begin{bmatrix} 1 & 0 & -1 & 3 \\ 1 & 0 & 3 & -2 \\ 2 & 0 & 2 & -1 \\ 1 & -3 & 1 & 2 \end{bmatrix} \quad A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$$

$$|A| = 0(C_{12}) + 0(C_{22}) + 0(C_{32}) + (-3)(C_{42})$$

$$C_{42} = \begin{bmatrix} 1 & -1 & 3 \\ 1 & 3 & -2 \\ 2 & 2 & -1 \end{bmatrix} = -8$$

$$|A| = (-3)(-8) = 24$$

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix}^T$$

$$C_{11} = \begin{bmatrix} 0 & 3 & -2 \\ 0 & 2 & -1 \\ -3 & 1 & 2 \end{bmatrix} = -3$$

$$C_{12} = - \begin{bmatrix} 1 & 3 & -2 \\ 2 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix} = 10$$

$$C_{13} = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 0 & -1 \\ 1 & -3 & 2 \end{bmatrix} = 9$$

$$C_{14} = - \begin{bmatrix} 1 & 0 & 3 \\ 2 & 0 & 2 \\ 1 & -3 & 1 \end{bmatrix} = 12$$

$$C_{21} = - \begin{bmatrix} 0 & -1 & 3 \\ 0 & 2 & -1 \\ -3 & 1 & 2 \end{bmatrix} = -15$$

$$C_{22} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix} = 10$$

$$C_{23} = - \begin{bmatrix} 1 & 0 & 3 \\ 2 & 0 & -1 \\ 1 & -3 & 2 \end{bmatrix} = 21$$

$$C_{24} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 2 \\ 1 & -3 & 1 \end{bmatrix} = 12$$

$$C_{31} = \begin{bmatrix} 0 & -1 & 3 \\ 0 & 2 & -2 \\ -3 & 1 & 2 \end{bmatrix} = 21$$

$$C_{32} = - \begin{bmatrix} 1 & -1 & 3 \\ 1 & 3 & -2 \\ 1 & 1 & 2 \end{bmatrix} = -6$$

$$C_{33} = \begin{bmatrix} 1 & 0 & 3 \\ 1 & 0 & -2 \\ 1 & -3 & 2 \end{bmatrix} = -15$$

$$C_{34} = - \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 3 \\ 1 & -3 & 1 \end{bmatrix} = -12$$

$$C_{41} = - \begin{bmatrix} 0 & -1 & 3 \\ 0 & 3 & -2 \\ 0 & 2 & -1 \end{bmatrix} = 0$$

$$C_{42} = \begin{bmatrix} 1 & -1 & 3 \\ 1 & 3 & -2 \\ 2 & 2 & -1 \end{bmatrix} = -8$$

$$C_{43} = - \begin{bmatrix} 1 & 0 & 3 \\ 1 & 0 & -2 \\ 2 & 0 & -1 \end{bmatrix} = 0$$

$$C_{44} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 3 \\ 2 & 0 & 2 \end{bmatrix} = 0$$

$$A^{-1} = \frac{1}{24} \cdot \left(\begin{bmatrix} -3 & 10 & 9 & 12 \\ -15 & 10 & 21 & 12 \\ 21 & -6 & -15 & -12 \\ 0 & -8 & 0 & 0 \end{bmatrix} \right)^T$$

$$= \frac{1}{24} \cdot \begin{bmatrix} -3 & -15 & 21 & 0 \\ 10 & 10 & -6 & -8 \\ 9 & 21 & -15 & 0 \\ 12 & 12 & -12 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{8} & -\frac{5}{8} & \frac{7}{8} & 0 \\ \frac{5}{12} & \frac{5}{12} & -\frac{1}{4} & -\frac{1}{3} \\ \frac{3}{8} & \frac{7}{8} & -\frac{5}{8} & 0 \\ \frac{1}{2} & \frac{1}{2} & -1 & 0 \end{bmatrix}$$

$$|A^{-1}| = (0)(C_{14}) + \left(-\frac{1}{3}\right)(C_{24}) + (0)(C_{34}) + (0)(C_{44})$$

$$= \left(-\frac{1}{3}\right) \begin{bmatrix} -\frac{1}{8} & -\frac{5}{8} & \frac{7}{8} \\ \frac{5}{12} & \frac{5}{12} & -\frac{1}{4} \\ \frac{3}{8} & \frac{7}{8} & -\frac{5}{8} \end{bmatrix}$$

$$= \frac{1}{24} \times$$